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4. The size of any population at a given time is determined by the equation:

Number of individuals = (birth rate + immigration) – (death rate + emigration)

In field studies which monitor population size over a period of time the number of individuals often stays constant.

(a) Using the terms in brackets from the above equation, write another equation which shows the relationship between the terms when the population size remains constant. [1]

.... = ...

Scientists monitored the population of frogs in a woodland surrounding a pond. The capture-mark-recapture method was used to determine the number of adult frogs, as follows:

- 19 frogs were caught
- marked by clipping off one toe
- they were then released back into the pond
- a week later the scientists collected as many frogs as they could over three consecutive days
- the results are shown in **Table 4.1**
- captured frogs from the three consecutive days were not released until after the third collection.

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Table 4.1 Result of collections following release of marked frogs

Date	Total no. of frogs captured	No. of marked frogs
Day 1	48	5
Day 2	45	5
Day 3	50	7
Total	143	17

(b) (i) From the figures given in the method and **Table 4.1** estimate the total number of frogs in the woodland, using the following formula: [2]

$$N = \frac{Mn}{m}$$

Where,

- N = number in population
- M = number initially captured and marked
- n = total number subsequently captured
- m = number of marked individuals recaptured.

Estimated number of frogs in the woodland = ...

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(ii) Explain why the chosen method of marking the frogs might have affected the estimate of the frog population. [1]

....

....

(c) Between capture and release the adult frogs were kept, ten to a tank, partially submerged in water collected from the pond. The frogs in one of the tanks developed red patches on their legs. The scientists suspected they were suffering from 'red-leg disease', caused by the bacterium, *Aeromonas hydrophila*, a Gram-negative bacillus.

The scientists took a swab from the leg of one of the frogs, performed a Gram stain and examined the sample under the microscope.

Describe the shape and colour of the bacteria they would have seen if the frog had been suffering from red-leg disease. [2]

Shape ....

Colour ....

(d) In order to study the survival rates of the larval stage of the frogs (tadpoles), two smaller ponds of equal volume were created from the existing pond using polyethylene sheets. Pond 1 was stocked with 5000 tadpoles per m^3 and pond 2 was stocked with 1000 tadpoles per m^3.

The scientists took 20 samples of water from each pond every ten days and counted the number of tadpoles in each sample. They used the mean counts to calculate the number of tadpoles per m^3. Their results are shown in **Graph 4.2**, the straight lines drawn in black are tangents to the curves.

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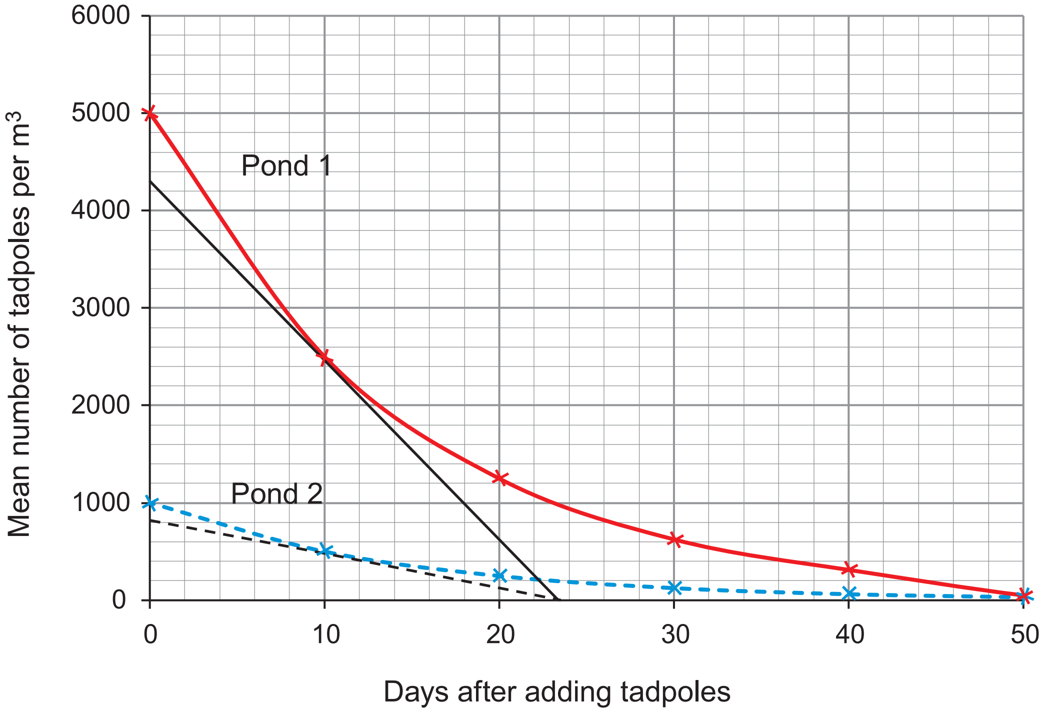
Graph 4.2



(i) Calculate the rate of decline in number of tadpoles for pond 1 **at day 10**. **Give your answer to two significant figures**. [3]

Rate of decline .... tadpoles m^{-3} day^{-1}

(ii) Using the information from **Graph 4.2** it was concluded that a density-dependent factor was causing the change in the tadpole populations. State the evidence for this conclusion and suggest what the factor might have been. [3]

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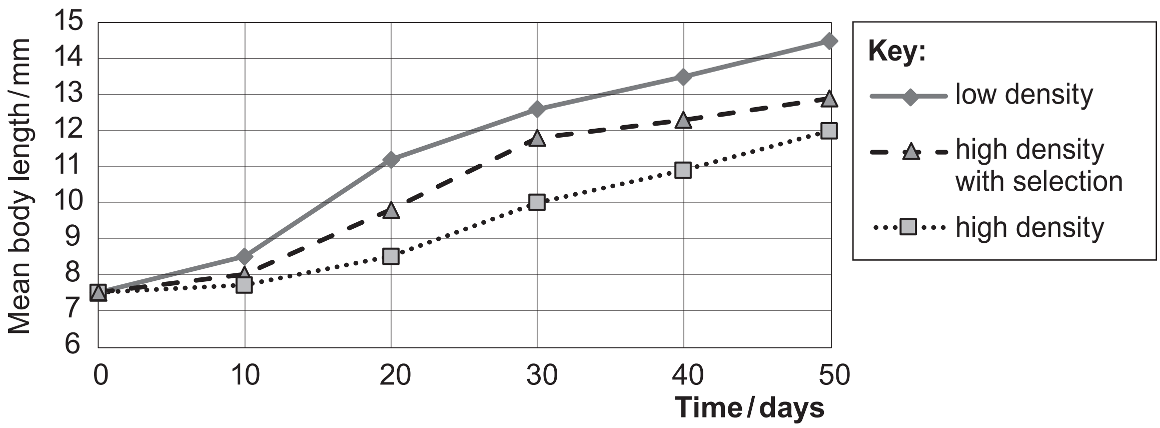
(e) The scientists also carried out a laboratory experiment to investigate the effect of selection on rate of development as measured by body length.
They used 3 different tanks, which contained 10 dm^3 of water.

- Tank 1 – low density (1 tadpole per dm^3)
- Tank 2 – high density (5 tadpoles per dm^3)
- Tank 3 – high density with selection against small individuals (two of the smallest tadpoles were removed each week).

All tanks were kept under the same environmental conditions.
The length of the tadpoles in each tank was measured every 10 days and a mean calculated. The results of the experiment are shown in **Graph 4.3**

Graph 4.3



(i) With reference to **Graph 4.3**, state **two** conclusions that can be drawn about the effect of density and selection on the rate of development of tadpoles in the three tanks. [2]

....

....

....

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(ii) Another group of scientists said that it was not valid to use the results from tank 2 and tank 3 to make a conclusion about the effect of selection. Suggest why it might not be valid to compare these two tanks **and** describe how the method could be changed to enable a more valid conclusion to be made. [2]

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